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Influencer Marketing Effectiveness

DIGITAL MARKETING INTELLIGENCE
ASSIGNMENT 2

STRATEGIC RECOMMENDATIONS FOR CAMPAIGN SUCCESS

EXECUTIVE SUMMARY

Messaging by Influencer Size

- Big influencers need bold emotions with few tags. Small ones thrive on authenticity and stories. Match tone to audience size.

Behind-the-Scenes Works

- Personal posts and thank-yours build connections, especially for luxury. Show real moments with the product.

Tag Smartly

- Utilize mentions and hashtags for visibility, BUT less might be more - over-tagging might appear overly commercial

Luxury Loves Emotion

- Use luxury-focused content and gratitude-based content for higher engagement.

Use Fear Wisely

- Sentiment fear boosts engagement. we could use these to spark discussions.

Don't Overdo Emotion

- More feeling isn't always better. Big influencers need balance to avoid seeming fake.

Final Tip

- Great campaigns mix luxury, smart emotion and perfect tagging - customized for each influencer. Guide, but don't control.

Expected		Unexpected
Strong Effect	As expected, the number of followers is the most important factor influencing comments and likes.	We expected positive emotions to affect engagement more, but it's also possible that people may find non-positive emotions more authentic.
Weak effect	The saturation of the images did not significantly affect engagement, as we suspected, it is not important compared to other variables.	The use of emojis was not proven to be very effective on boosting engagement



@MARALAFONTAN



Recommendations resulting from data limitations of this project:

- Increase the sample size
- Select the appropriate web celebrities according to the product positioning, e.g., personalized web celebrities for personalized products, and mega web celebrities for luxury perfumes.
- Control the difference in the number of followers in the sample selection to minimize the great influence of the number of followers on the significance of the other variables in the model.

CHOSEN INFLUENCERS

@BADGALRIRI

Literature based

- Older figures (>31) were found to be more influential on average (Aral & Walker, 2012) - 4/5 influencers we chose are older or around 31 years old
- Lower following base --> higher engagement & lower cost per endorsement (Wies et al., 2022; Beichert et al., 2024) - Alexandra Saint Mleux and Mara LaFontan < 2M followers

@BELLAHADID

@REALBARBARAPALIN

Aesthetic Based

- **Rihanna** - Massive Global Influence (the fourth most influential celebrity in America) (Staff, 2023). - Association with Rihanna would not only bring a massive audience reach but also instant credibility for the brand due to her expertise in the Beauty and Fragrance Industry
- **Bella Hadid** - Top-tier fashion model; Victoria's Secret Angel; known trend-setter, thus an ideal choice for perfume branding
- **Barbara Palvin** - High Fashion and Editorial Appeal - Victoria's Secret Angel, thus her image is strongly linked to elegance, femininity and sensuality
- **Mara LaFontan** - Parisian Chic and Romantic Aesthetic - her content embodies effortless elegance and she is highly engaged with her followers
- **Alexandra Saint Mleux** - Luxury Aesthetic, Beauty, and Artistic Appeal - rising influencer known for her elegant presence, with content blending art, fashion, skincare, and vintage aesthetics.

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@ALEXANDRASAINTMLEUX

DATASET OVERVIEW

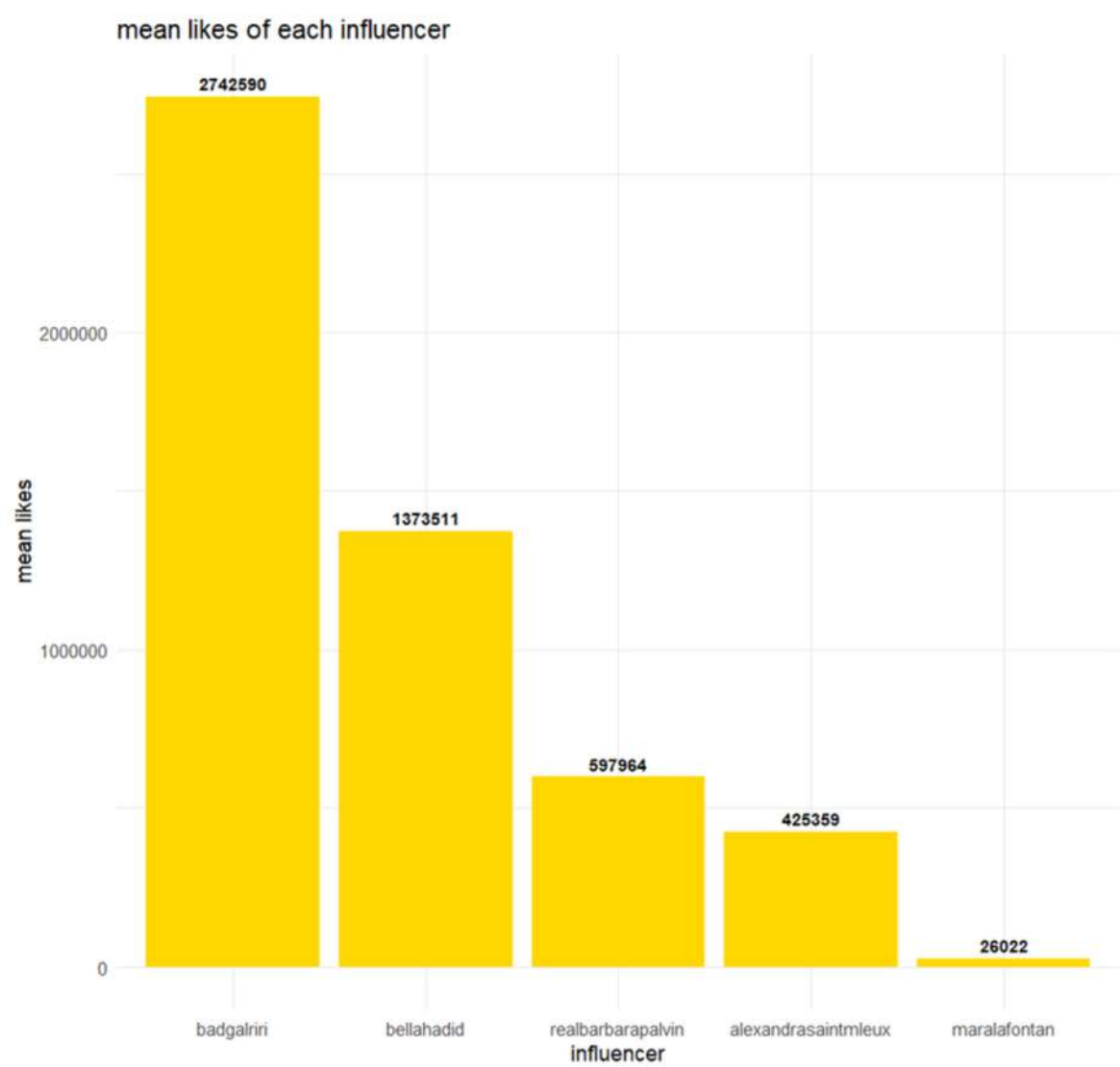
We collected 250 Instagram posts (50 per influencer) along with:

- Post-level data: likes, comments, captions, emojis, timestamps.
- Image-level data: brightness, saturation, object detection, etc.
- Influencer bios: followers, following, post counts.

LIMITATIONS

- The effect of other variables on engagement becomes insignificant due to the large difference in the amount of fans of the selected web celebrities.
- Due to the uncertainty of the type of perfume and the lack of understanding of the positioning and target group of the perfume, there is a lack of targeting in the selection of web celebrities.
- The number of observations is too small, the number of netizens and the number of posts per netizen should be increased.

HOW ENGAGING ARE OUR CHOSEN INFLUENCERS?



Influencer Profile Statistics:

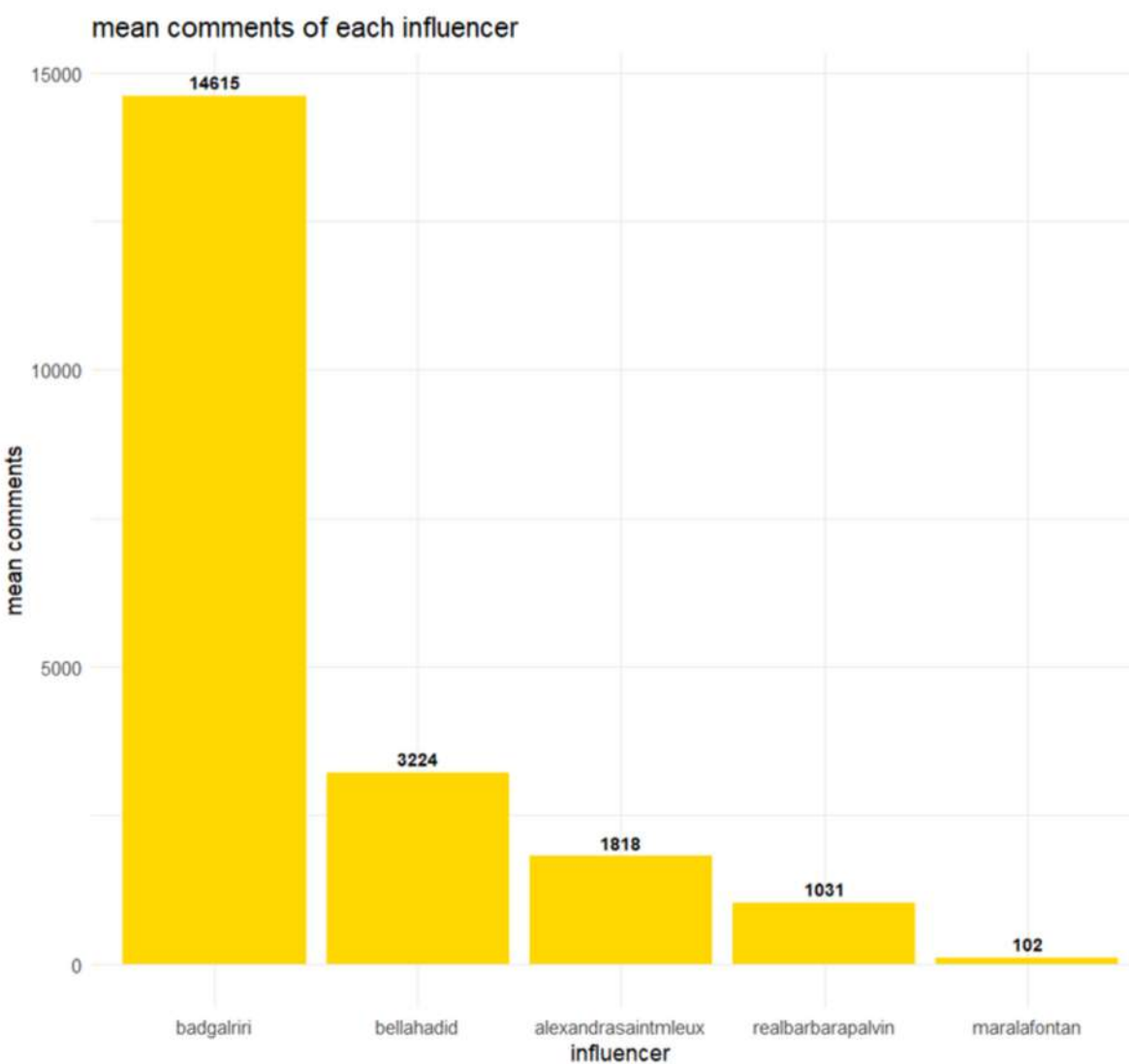
- @badgalriri: 149M followers, 1.6K following, 4.9K posts
- @bellahadid: 61.2M followers, 989 following, 851 posts
- @realbarbarapalvin: 20.8M followers, 370 following, 1.9K posts
- @maralafontan: 1.1M followers, 1K following, 1.5K posts
- @alexandrasaintmleux: 1.6M followers, 933 following, 108 posts

FIGURE_ MEAN LIKES FOR EACH INFLUENCER

@BADGALRIRI LEADS WITH OVER 2.7M LIKES PER POST.

FIGURE_ MEAN COMMENTS FOR EACH INFLUENCER

AGAIN, @BADGALRIRI RECEIVES AN AVERAGE OF 14.6K COMMENTS PER POST – SIGNIFICANTLY ABOVE THE REST.



WHAT DO INFLUENCERS TALK ABOUT?

Caption Lengths & Emoji Usage:

Captions vary widely in length, reflecting different communication styles across influencers.

Average Number of Words per Caption:

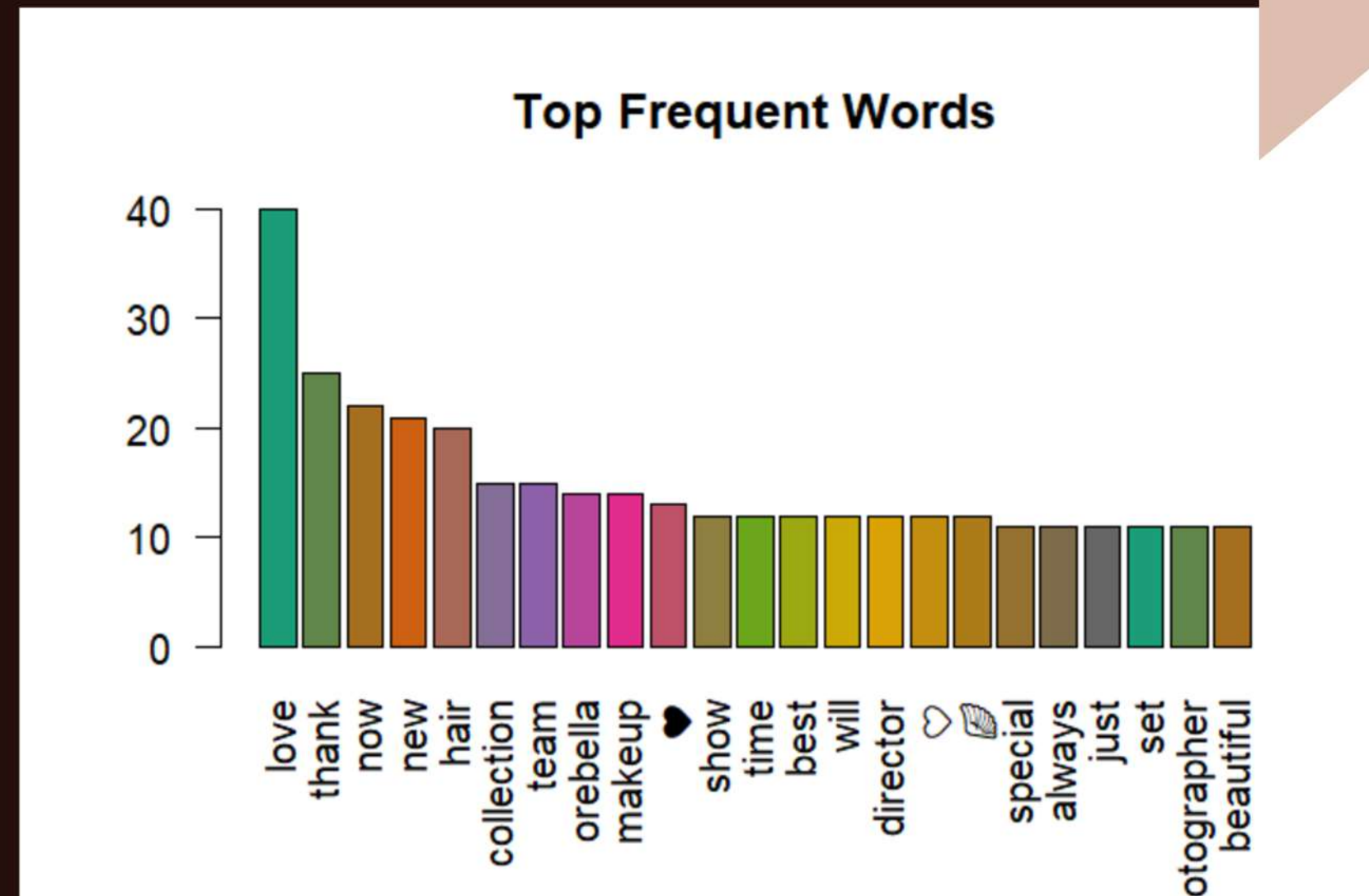
- @bellahadid: 48.6 (most detailed)
- @badgalriri: 18.7
- @realbarbarapalvin: 17.9
- @alexandrasaintmleux: 10.7
- @maralafontan: 9.3 (shortest)

• Average Emoji Use:

- Emojis per post range from 0.6 to 1.8, with @alexandrasaintmleux using the most.
- Emojis often carry tone and emotions which is a key factor in expressive posts.



FIGURE_ TOP FREQUENT WORDS



Top Words in Captions:

Posts often express emotion, gratitude, and creativity.

Most frequent words include:

“love”, “thank”, “now”, “new”, “hair”, “collection”, “team”, “orebella”, “makeup”, “show”.

These reflect themes of appreciation, beauty, events, and product promotion – highly relevant to perfume branding.

SENTIMENT AND TEST-BASED FEATURE EXTRACTION

Bar Charts per Influencer

- Barbara Palvin has the highest mean and median likes and comments, signaling consistent engagement with her audience
- Bella Hadid also performs well, but in this case the variance is higher, thus showing lower consistency

These insights show that for audience engagement it is better to go for these two influencers , while the others could provide other valuable benefits for the campaign such as high visibility

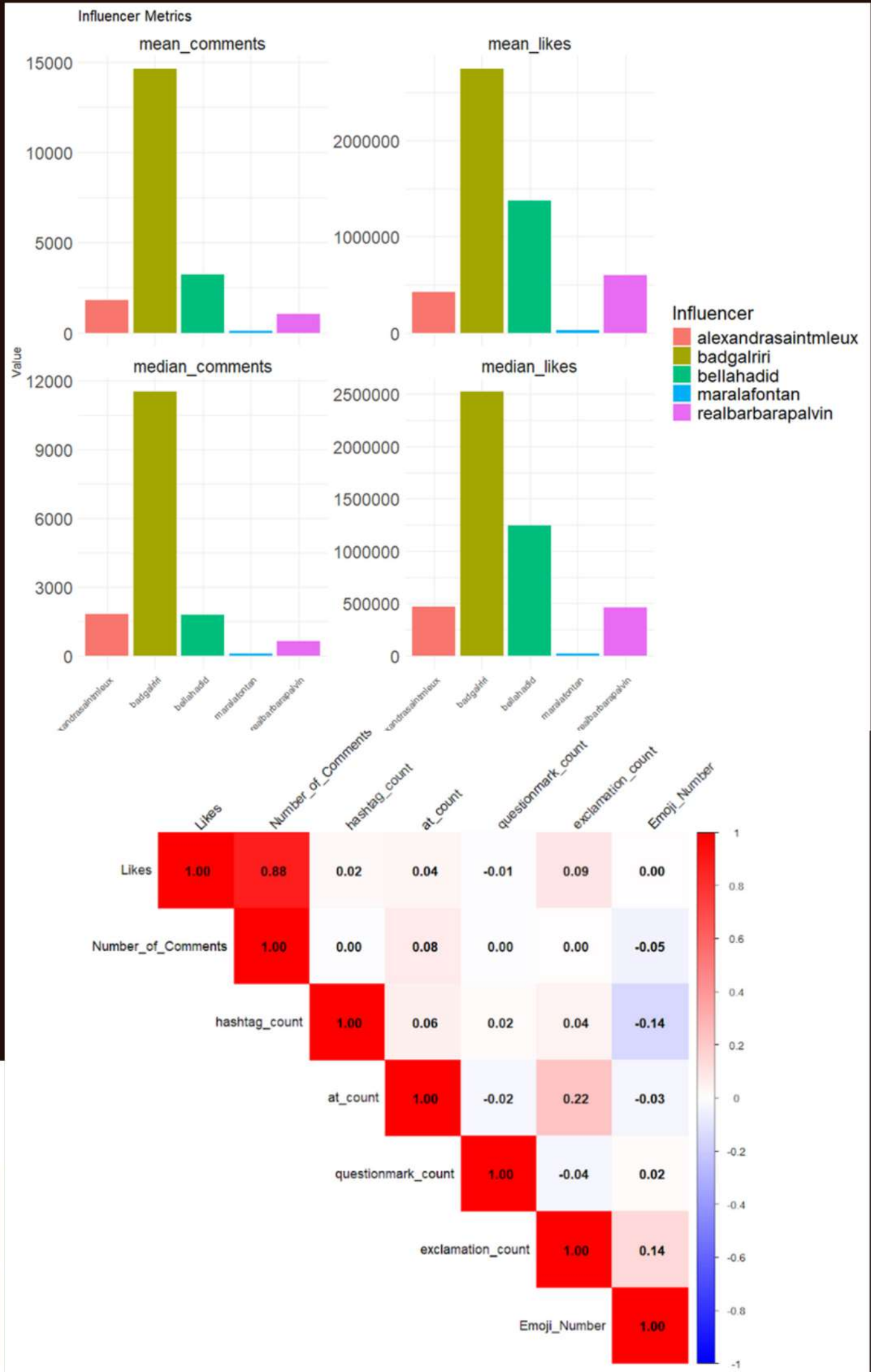
Correlation Heatmap:

- The likes and comments are strongly correlated (1.00 and .98)
- Hashtags are also positively correlated to Likes (.68) and Comments (.60). Thus more hashtags result in more engagement as the audience can discover the content better (Tjandra et al., 2023)
- Punctuation Count and Emoji Number are showing lower correlations (.42 and .14), suggesting that the extent to which the caption was more expressive or contained more emojis does not greatly affect the engagement. Literature has shown that emojis have a high impact on engagement mostly in combination with positive emotions (Ko et al., 2022). However, we discovered that in our sample, the negative sentiments were more significant.

Significant Predictors

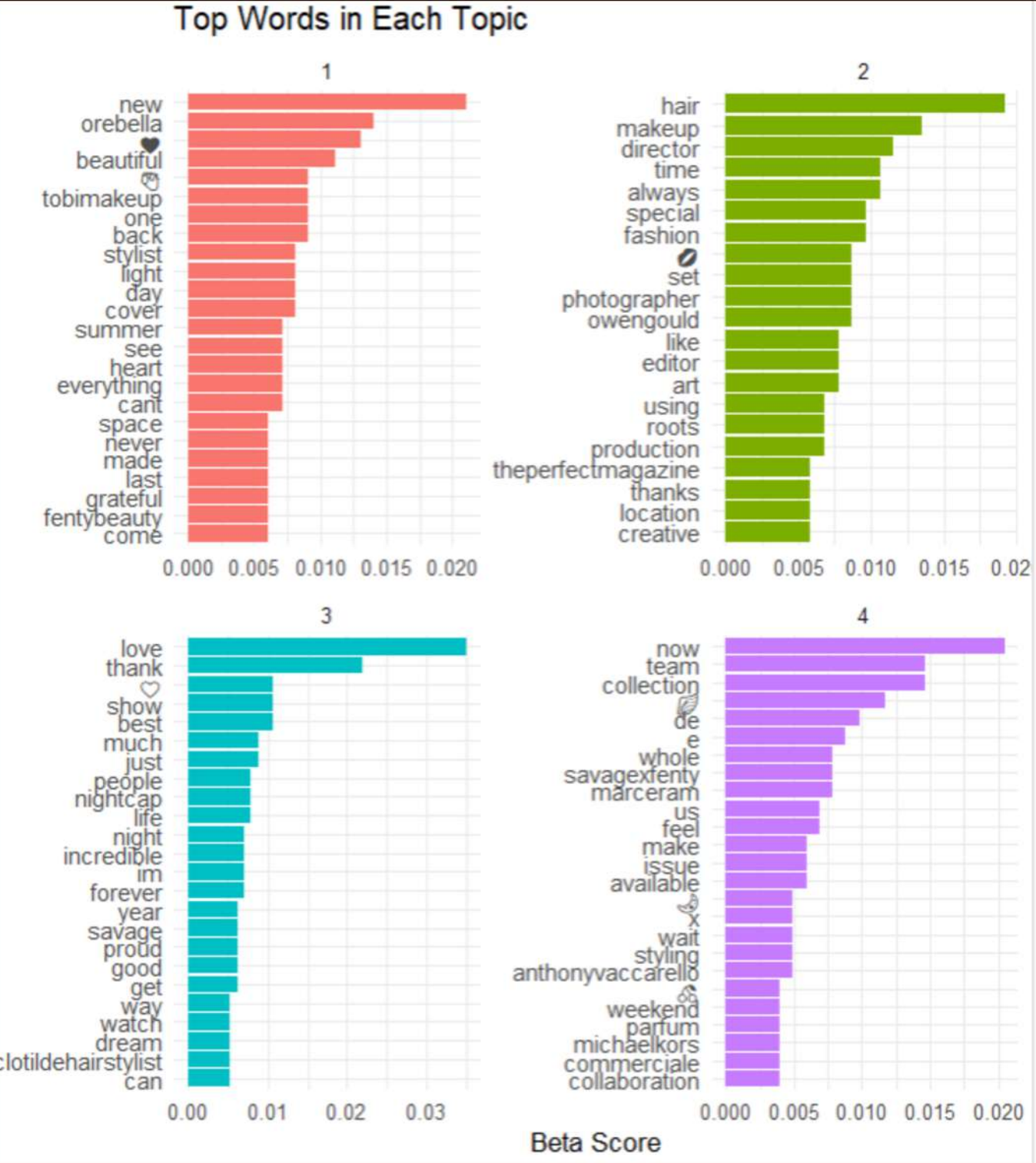
- Fear was significant and showed positive effect in relation to comments. Thus fear expressions could increase the number of comments
- Disgust was also significant and positive, meaning that it could also influence the number of comments

Variable	Estimate	P-value
fear	2716.06	.001**
disgust	3295.35	.010*

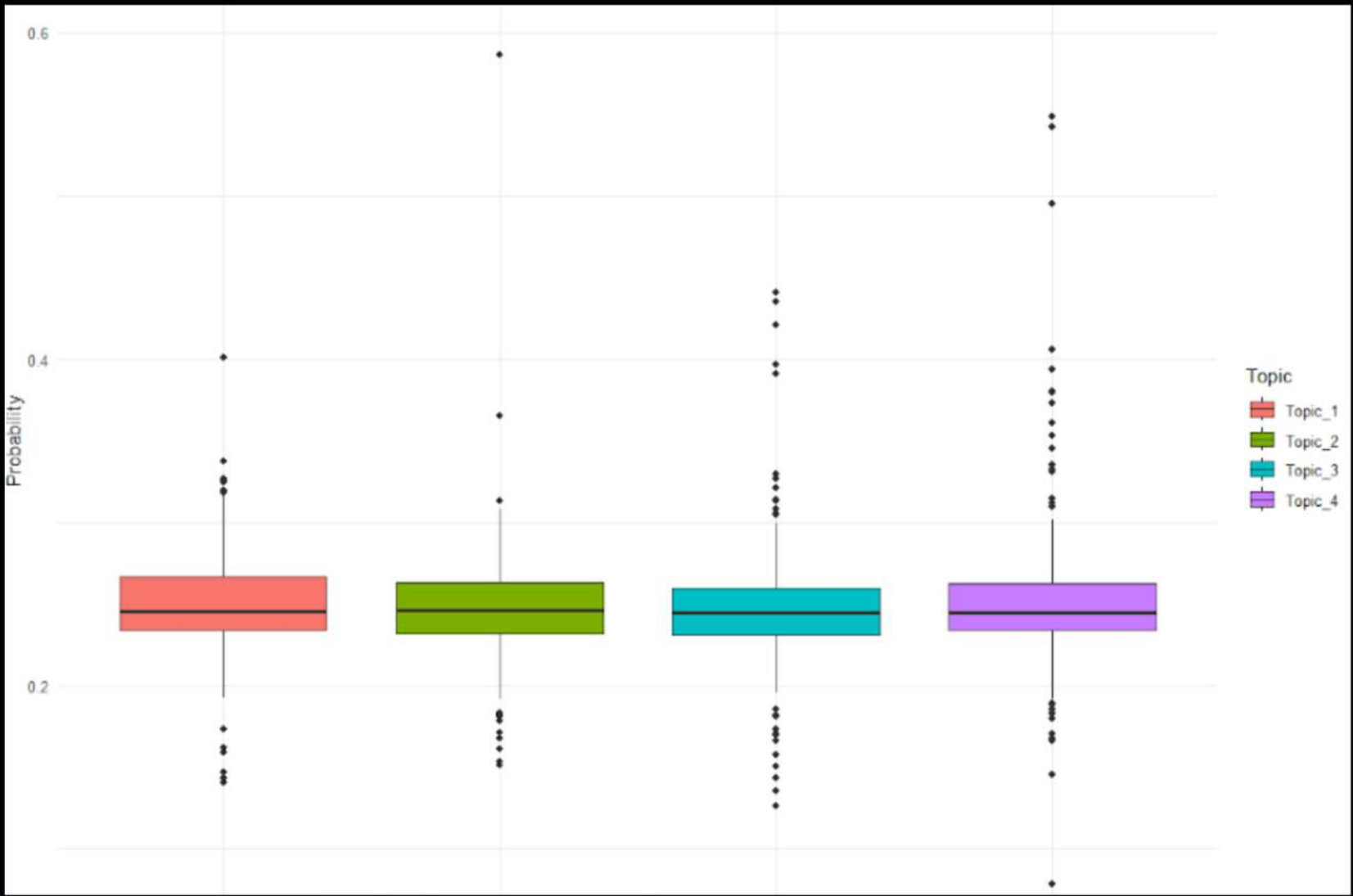


TOPIC DISCOVERY IN CAPTIONS

Optimal topics: 4 (selected via metrics).
Topics 1 : red: = luxury
Topics 2: green = seasonal beauty trends
Topics 3: blue = gratitude
Topics 4: purple = behind-the-scenes



FIGURE_ TOP WORDS OF EACH TOPIC



The relatively balanced probability across topics can mean that variation in engagement is less likely to be a cause of imbalanced topic frequency but more likely due to inherent differences in the audience response based on different themes. However, despite the even distribution, engagement varied across words in certain topics. For example, the posts dominated by gratitude showed a lower engagement compared to the ones dominated by luxury. The analysis suggested that the most efficient strategy could be balancing content between themes to reach a higher audience.

WHAT PREDICTS LIKES AND COMMENTS?

Model Type	Purpose	Adjusted R ²
Stepwise Regression – Likes	Interpret key like drivers	0.5824
Stepwise Regression – Comments	Interpret key comment drivers	0.6093
Random Forest – Likes	Validate variable importance	0.537
Random Forest – Comments	Validate variable importance	0.524

To conclude:

- Followers are foundational for both likes and comments
- Emotion matters: Positive emotions like joy, trust, and excitement are strong engagement boosters.
- Visuals amplify textual impact: brightness, saturation, and object richness boost both metrics.
- Mentions and emojis subtly boost relatability and engagement.

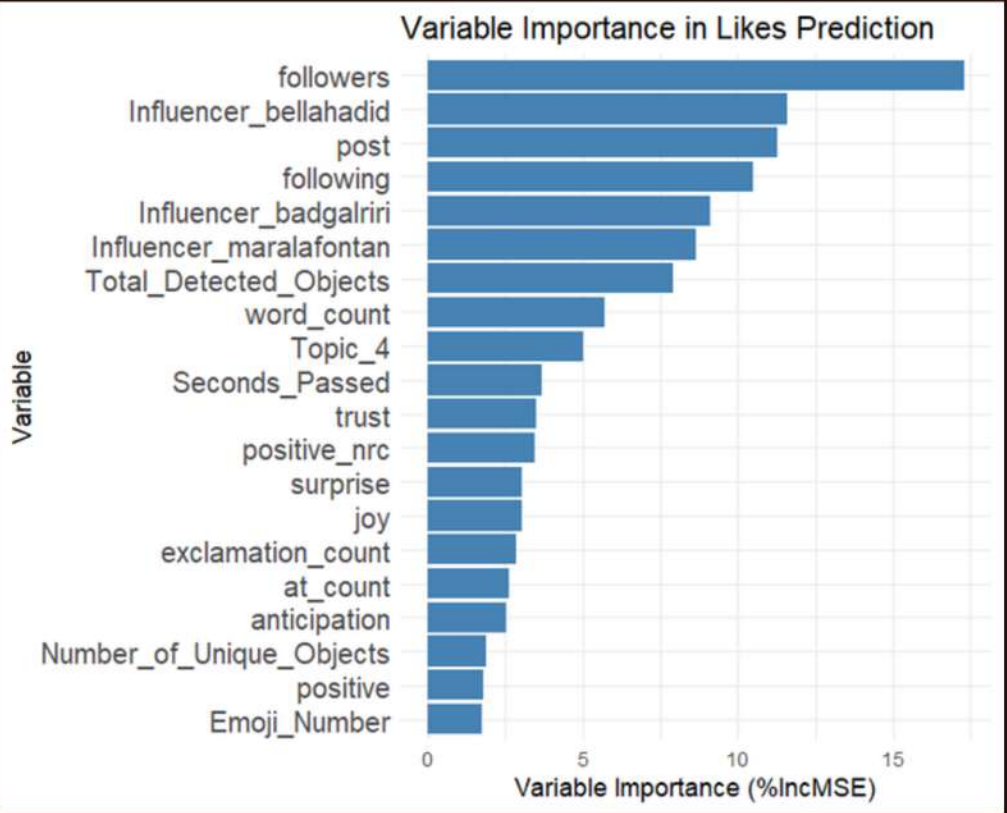


Top Drivers of Likes

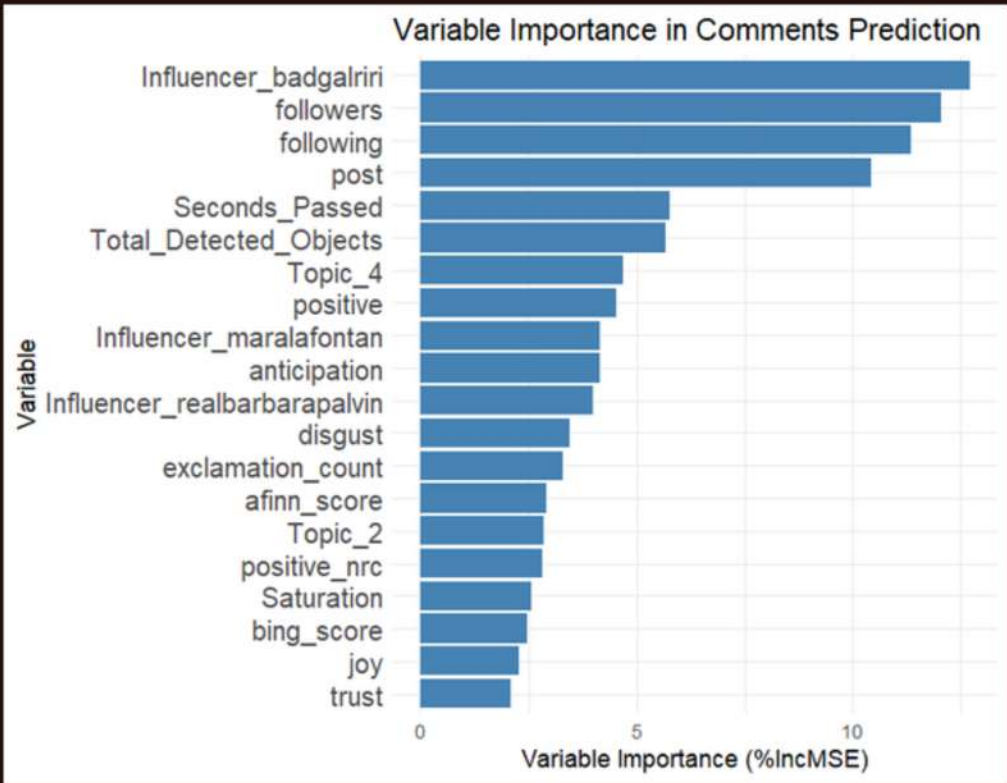
From both modeling approaches, the following stood out:

- Followers: Larger reach = more likes
- Positive emotions: Joy and Trust were associated with higher like counts – audiences respond to uplifting, authentic messaging.
- Topic_4: Posts in the "behind-the-scenes" cluster get more likes.
- @Mentions: Tagging others boosts reach and signals collaboration.

FIGURE_ VARIABLE IMPORTANCE IN LIKES PREDICTION



FIGURE_ VARIABLE IMPORTANCE IN COMMENTS PREDICTION



Top Drivers of Comments

Key predictors include:

- Followers: More reach drives more interaction – especially in the mid-tier range.
- Brightness: Visually engaging images spark conversation.
- Joy, Surprise, and Anticipation: Positive anticipation or excitement (e.g., “can’t wait,” “so excited”) increases interactions.
- Emoji Use & Mentions: Playful, expressive captions invite conversation.

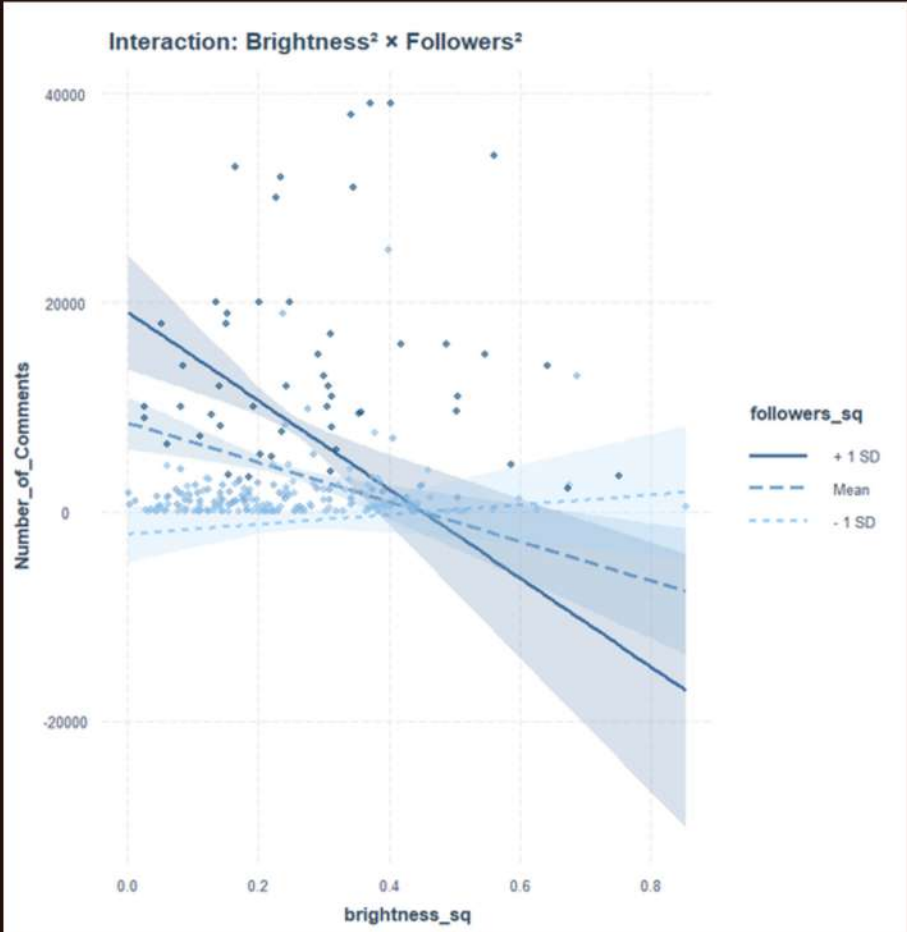
NON-LINEAR & INTERACTION EFFECTS ON ENGAGEMENT

Aspect	Likes Model	Comments Model
Univariable Effects	followers, at_count remained significant even after controlling for nonlinear terms. fear ² showed a positive quadratic relationship with Likes, indicating that moderate levels of fear expressions might reduce Likes, but extreme levels can increase them.	followers ² showed a quadratic relationship with comments, suggesting diminishing or increasing effects depending on the range.
Interaction Effect	- followers × anger (Strengthening effect) - followers × negative_nrc² (Weakening effect)	- followers² × Brightness (Strengthening effect) - followers² × anger (Strengthening effect) - followers² × negative_nrc² (Strengthening effect) - Seconds_Passed² × disgust (Weakening effect)
Effect size	Moderate to strong effects for followers and interaction terms, enhancing the model's predictive ability.	Most effects are statistically significant but relatively small in magnitude. Interactions are more subtle.
R ² Improvement	Increased by 5% (from 0.58 to 0.63), indicating including non-linear and interaction patterns enhance the explanatory power of a model .	Increased by 12% (from ~0.60 to 0.72), improving fit though explanatory power is more spread out across variables.

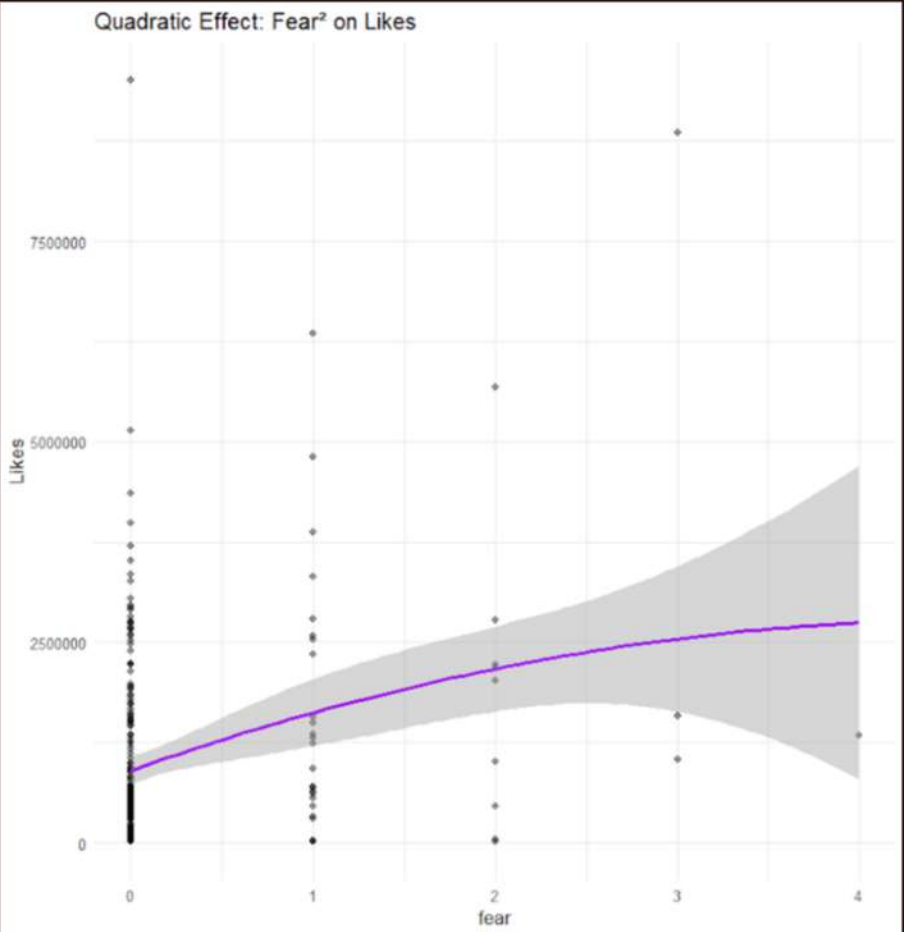
TO CONCLUDE:

- ENGAGEMENT IS NOT LINEAR – SOME EFFECTS INCREASE AND THEN TAPER OFF, WHILE OTHERS REVERSE BEYOND A CERTAIN POINT.
- NEGATIVE EMOTIONAL TONES (E.G. ANGER, NEGATIVITY, DISGUST) FOR MID-SIZED INFLUENCERS, THEY CAN DRIVE ENGAGEMENT BY SPARKING CONVERSATIONS UNDER CERTAIN CONDITIONS. FOR LARGE INFLUENCERS, HOWEVER, THE SAME TONE CAN BACKFIRE–LEADING TO DRAMATIC DROPS IN COMMENTS OR NEGATIVE SENTIMENT. IF A PERFUME CAMPAIGN IS INTERPRETED AS TOO AGGRESSIVE, OFF-TONE, OR OVERLY CRITICAL, THE REPUTATIONAL IMPACT WOULD BE SIGNIFICANT.
- THE KEY IS TO BE STRATEGIC WITH EMOTIONAL APPROACH, USING POSITIVITY, ELEGANCE AND INSPIRATION – ESPECIALLY FOR INFLUENCERS WITH A LARGE REACH.

FIGURE_ NUMBER INTERACTION EFFECT
BRIGHTNESS² × FOLLOWERS² COMMENTS



FIGURE_ NUMBER QUADRATIC
RELATIONSHIP BETWEEN FEAR AND LIKES



Thank You

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